RF-STAR REFLECTS ISOMORPHISMS FOR AFFINE MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 972-982, immediately following the statement at 964-970.

Editorial note. The first sentence of the proof unpacks what 'reflects isomorphisms' means; that reading is part of the counted span.

Allowed standard prerequisites (editorial).

- The scheme case. Derived Categories of Schemes, Lemma perfect-lemma-affine-morphism, cited at proof lines 978-979. Terminal input: Rf_* reflects isomorphisms for affine morphisms of schemes.
- The etale-site equivalence for D_QCoh. Lemma lemma-derived-quasi-coherent-small-etale-site of this chapter, cited at proof lines 979-981. Identifies D_QCoh(O_X) with D_QCoh of the small etale site via Repsilon_*.
- Comparison of total direct images. Remark remark-match-total-direct-images of this chapter, cited at proof lines 979-981. Matches Rf_* with the scheme-side functor under the equivalence, so the scheme result transfers.

Source reference. Original lemma 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08II: focused-seeds/geometry/S9-perf-affine-morphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08GH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 896–896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over S. Then $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ reflects isomorphisms.*

Proof. The statement means that a morphism $\alpha : E \to F$ of $D_{QCoh}(\mathcal{O}_X)$ is an isomorphism if $Rf_*\alpha$ is an isomorphism. We may check this on cohomology sheaves.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

In particular, the question is étale local on Y. Hence we may assume Y and therefore X is affine. In this case the problem reduces to the case of schemes (Derived Categories of Schemes, Lemma 08II) via Lemma 071Q and Remark 08GH. $\square$